East West University

Dept. of Mathematical and Physical Sciences (MPS)
Course Instructor: Afsana Al Sharmin

Time: 90minutes **Full marks: 30**

Answer all questions
Marks

1. In a class of 50 students, 28 like to study genetics, 22 like to study microbiology, and 12 like both subjects. If a student is chosen randomly, then: 3+2

   (i) What is the probability that the student likes neither genetics nor microbiology?
   (ii) Are the events "like genetics" and "likes microbiology" mutually exclusive? Explain your answer.

2. In a plant genetics lab, there are 15 seedlings. The probability that a seedling carries a recessive gene for a disease is 0.25. If 5 seedlings are selected at random, 3+1

   (i) What is the probability that at least 2 seedlings carry the recessive gene?
   (ii) What is the average number of seedlings expected to carry the gene in this selection?

3. What is the level of significance and type two error? Suppose you measured the fold-change of a gene in a sample of 10 independent replicates. The observed sample mean was 2.30 with a sample standard deviation of 0.80. At the 1% significance level, is the mean fold-change significantly different from 1.0? 1+4

4. A researcher is studying the relationship between **plant type** and **resistance to a particular disease**. The following data is collected from 100 plants. Is there a significant association between plant type and disease resistance? 5+3

Plant Type	**Resistant**	**Not Resistant**	**Total**
Genetically Modified (GM)	30	20	50
Non-GM	10	40	50
Total	**40**	**60**	**100**

   (i) Is there a significant association between plant type and disease resistance?
   (ii) What is the probability that the plant is GM, given that it is Resistant?

5.	(i)The body temperature of healthy rats is normally distributed with a mean of 37.5°C and a standard deviation of 0.5°C. What is the probability that a randomly selected rat has a body temperature between 37°C and 38°C? (ii)What is level of significance? Write the properties of Normal distribution? (iii)what is type one and type two ANOVA? explain it with an example.	3 1+2 2

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